Challenge 9: Union-Closed Sets Conjecture

Definition 1 (Union-closed set)**.** A set $\mathcal{F}$ of sets is *union-closed* if for any two sets in $\mathcal{F}$, their union is also in $\mathcal{F}$. For a nonempty finite set , the *density* of an element x with respect to is defined to be

$$\frac{|\{F \in \mathcal{F} : x \in F\}|}{||}$$

Conjecture 1.1 (Union-closed sets conjecture)**.** For every finite union-closed set $\mathcal{F} \notin \{\emptyset, \{\emptyset\}\}$, there exists an element x of density at least one half.

We are interested in the following parametrized version of the conjecture.

Conjecture 1.2 (Scaled union-closed sets conjecture)**.** Let $r \in \mathbb{N}_{>2}$. For every finite union-closed set $\mathcal{F} \notin \{\emptyset, \{\emptyset\}\}$, there exists an element x of density at least $\frac{1}{2} - \frac{1}{r}$.

The above conjecture is open for all values of $r > 2$.